

Title: Design and Simulation of a Comprehensive Relay-Based Protection Scheme Using ETAP

System Description with diagram:

In our one line diagram we have used a power grid which is (1398 MVA_{sc} with XR ratio of 14) in 3 phase and is in swing. We have used two 33 KV high voltage side buses named HAMDA and CAMPUS. From the bus named CAMPUS two branches come out. Each branch has one transformer (33/11KV, 10.66 impedance, XR ratio of 20), one low voltage bus of 11 KV, and one load named SCHOOL(2.5MVA) and HOSPITAL(3MVA). We also have a transmission line made of Aluminum which is about 2 KM named GROSBEAK connected between two 33KV buses. We have used total of 7 CT, Relay switches (over current, over voltage) and High voltage Circuit Breakers.

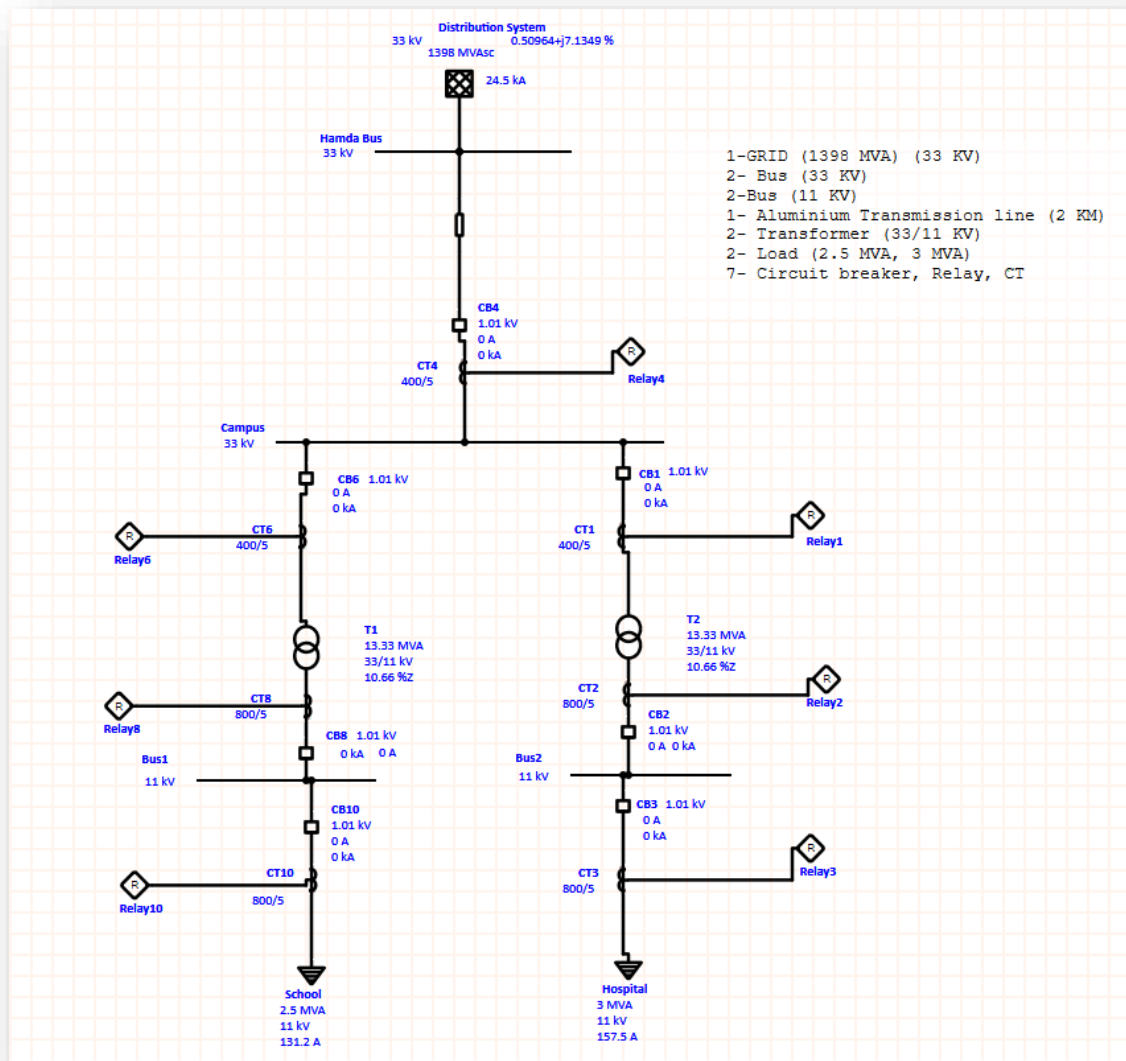


Fig 1: Circuit diagram in Normal mode.

Load flow analysis:

We can see from the diagram that the percentage of voltage across the buses are more then 95% which means that all the buses are working perfectly under the load flow. We can clearly ensure it by seeing the current flow as the high voltage side are having lower current and also for the long transmission line little amount of current dropped from 95.2 A to one branch having 51.9A another one having 43.3 A. At the low voltage side currents are 130 and 155.8 A according to the branch requirement.

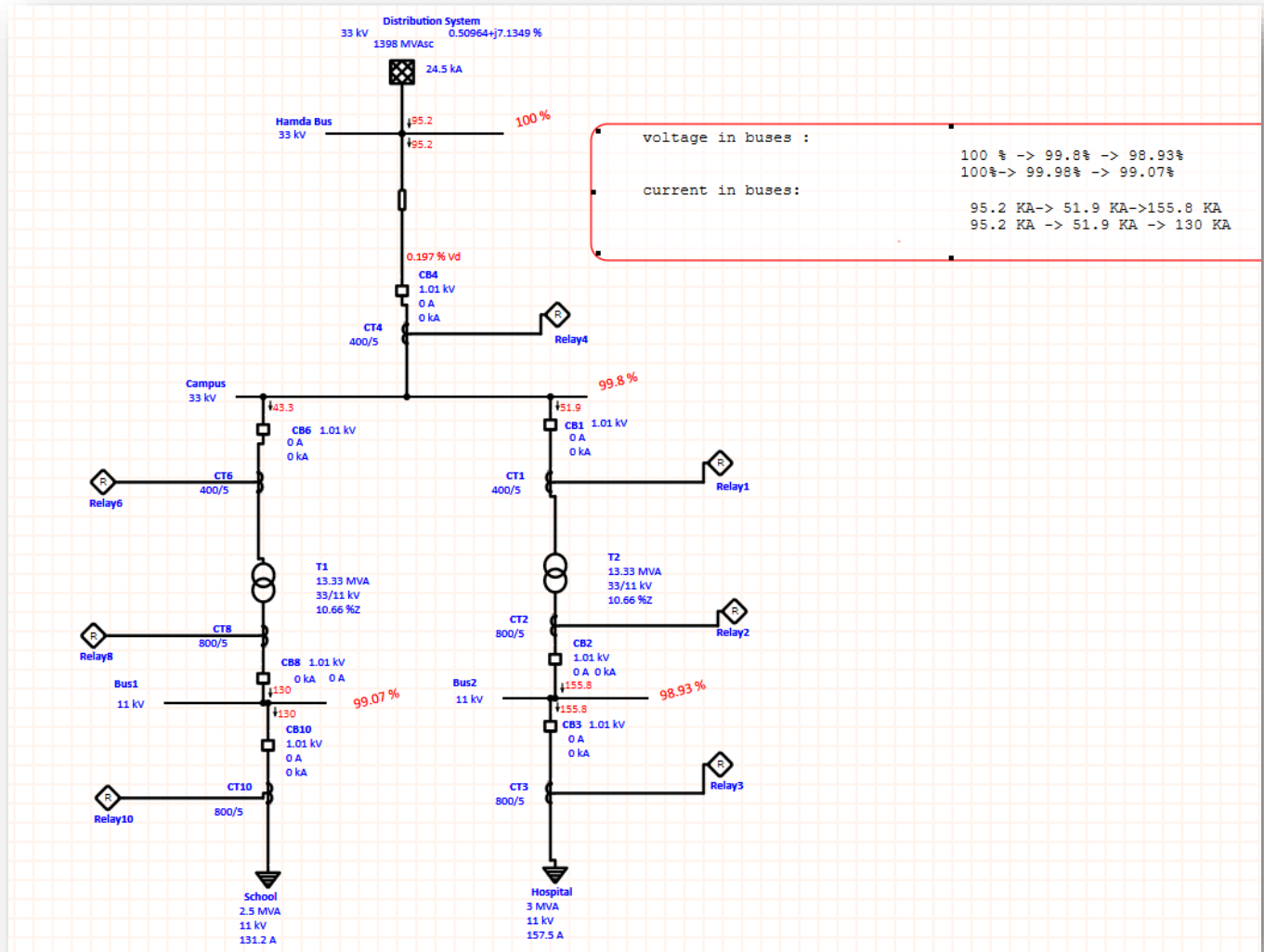


Fig 2: Load flow analysis

Protection Scheme design with calculation:

In our system we have used 7 CT with relay and circuit breaker. The CT that are connected to the high voltage side of the transformer or 33 KV bus are basically of 400:5 ratio with a burden of 30VA. But the CT that we used on the low voltage side of the transformer or 11 KV bus are of 800:5 ratio having burden of 30 VA. The CTs are different because when voltage goes down, current goes up.

- HV (33kV): High Voltage → Low Current → Smaller CT (400:5)
- LV (13kV): Low Voltage → High Current → Larger CT (800:5)

Each circuit breaker are of 1.01 KV with the delay of 0.01s so that it can react according to the timing of the relay. Now to choose the relay for high voltage and low voltage side of the diagram we had to go through some calculations.

As, the plug setting usually ranges from 50% to 200% so we took it as 130% or 1.3. considering it we get our

$$\text{pickup current} = \text{full load current} * 1.3$$

$$\text{full load current} = \frac{\text{transformer MVA}}{\sqrt{3} * \text{voltage KV}}$$

$$\text{now, High Voltage full load current} = \frac{10}{\sqrt{3} * 33} = 174 \text{ A}$$

$$\text{Low Voltage full load current} = \frac{10}{\sqrt{3} * 11} = 524 \text{ A}$$

$$\text{High Voltage pickup current} = 174 * 1.3 = 226.2 \text{ A}$$

$$\text{Low Voltage pickup current} = 524 * 1.3 = 681.2 \text{ A}$$

Now to set up relay we assumed the delay time for relay will be 0.3 means after each 0.3s another relay will respond to a fault. We assumed the operating time of the base relay is $t = 0.5\text{s}$

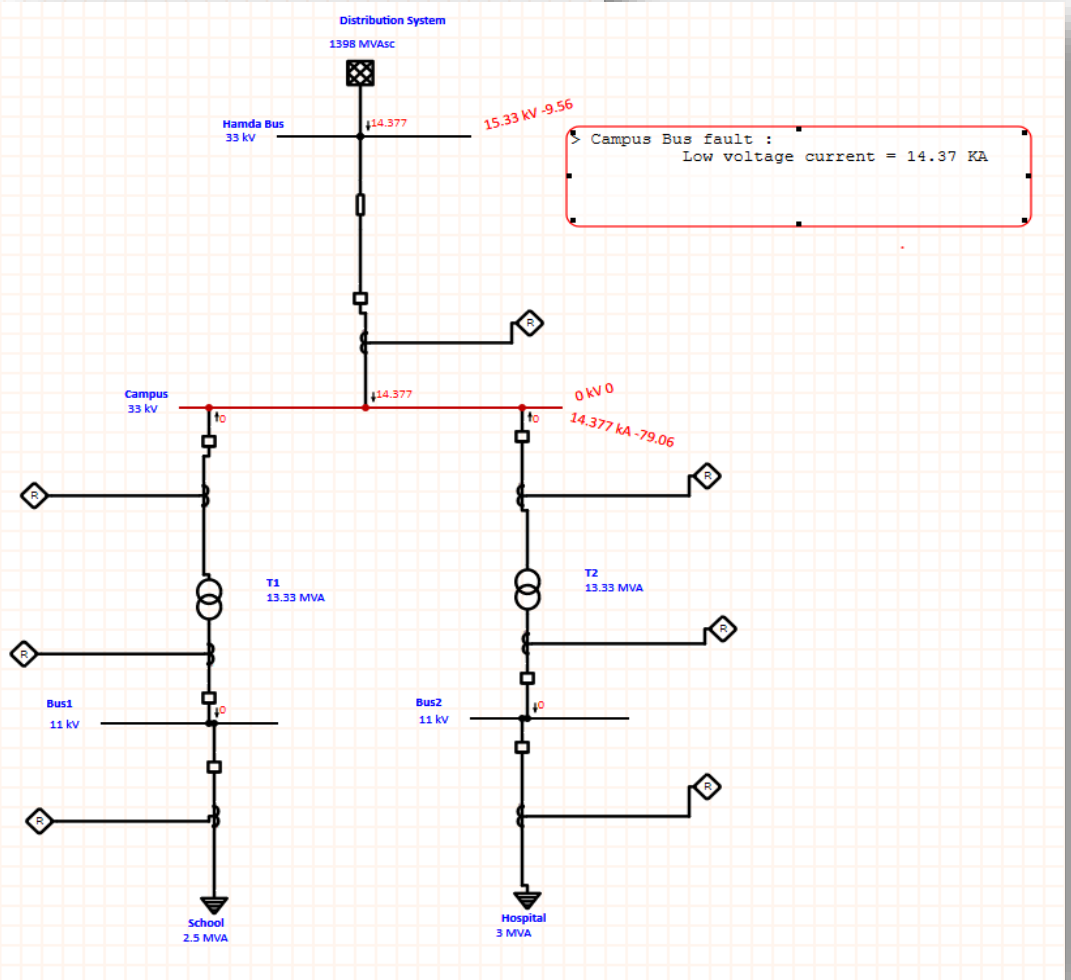
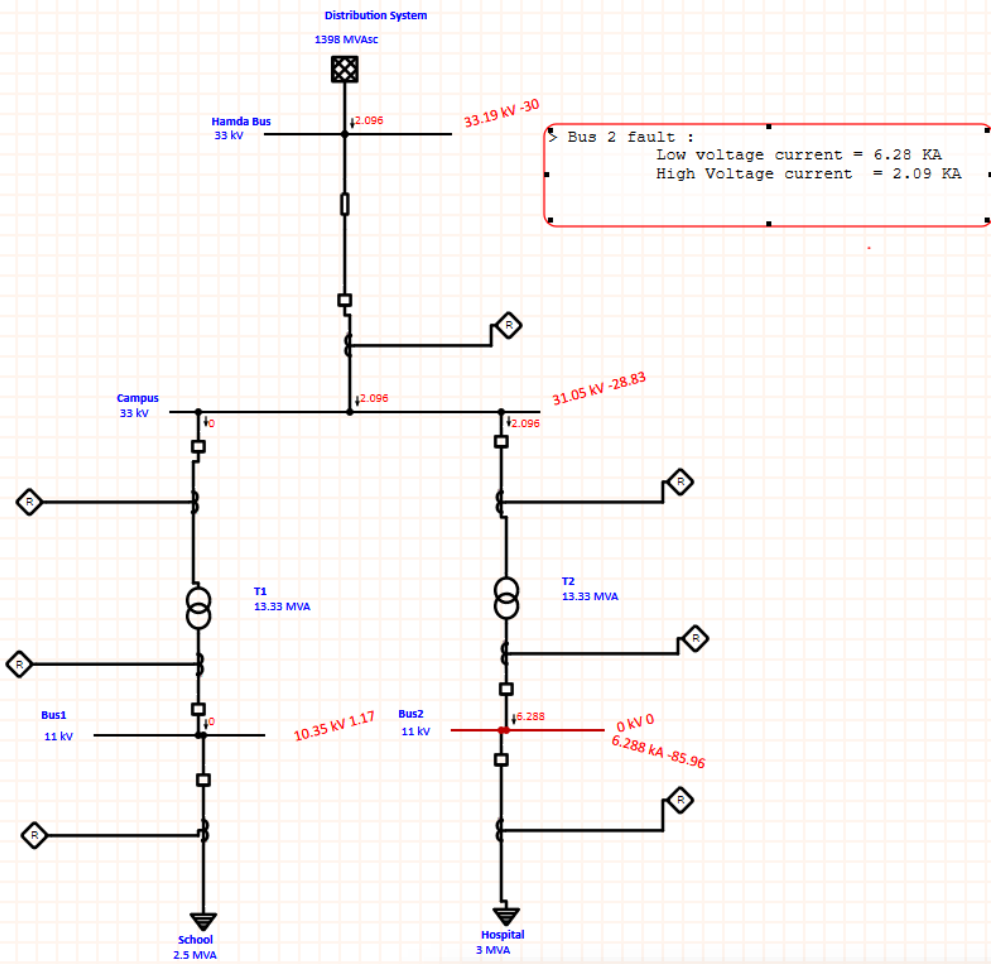
So our calculation goes as follows:

$$t = \frac{TSM * 0.14}{PSM^{0.02} - 1}, TSM = \frac{t * (PSM^{0.02} - 1)}{0.14}$$

$$PSM = \frac{\text{fault current}}{\text{pick up current}}$$

Fault current generation from the diagram:

If we consider the load side and low voltage side of bus is at fault in the diagram, we get Low Voltage fault current 6.28 KA and High Voltage fault current 2.09 KA and If we consider the transmission line side and high voltage side of bus is at fault in the diagram we get High Voltage fault current 2.09 KA.



Relay setting and selection:

For the basic relay connected to the load side (relay 3, 10) calculation goes:

$t = 0.5s$ here, fault current = 6.28 KA, pickup current = 681.2 A

PSM = 9.21

TSM = 0.162 s

For the relay connected to the transformer low voltage side (relay 2, 8) calculation goes:

$t = (0.5 + 0.3) = 0.8s$ here, fault current = 2.09 KA, pickup current = 681.2 A

PSM = 3.068

TSM = 0.26 s

For the relay connected to the transformer high voltage side (relay 1, 6) calculation goes:

$t = (0.8 + 0.3) = 1.1 s$ here, fault current = 14.37 KA, pickup current = 226.2 A

PSM = 63.52

TSM = 0.68 s

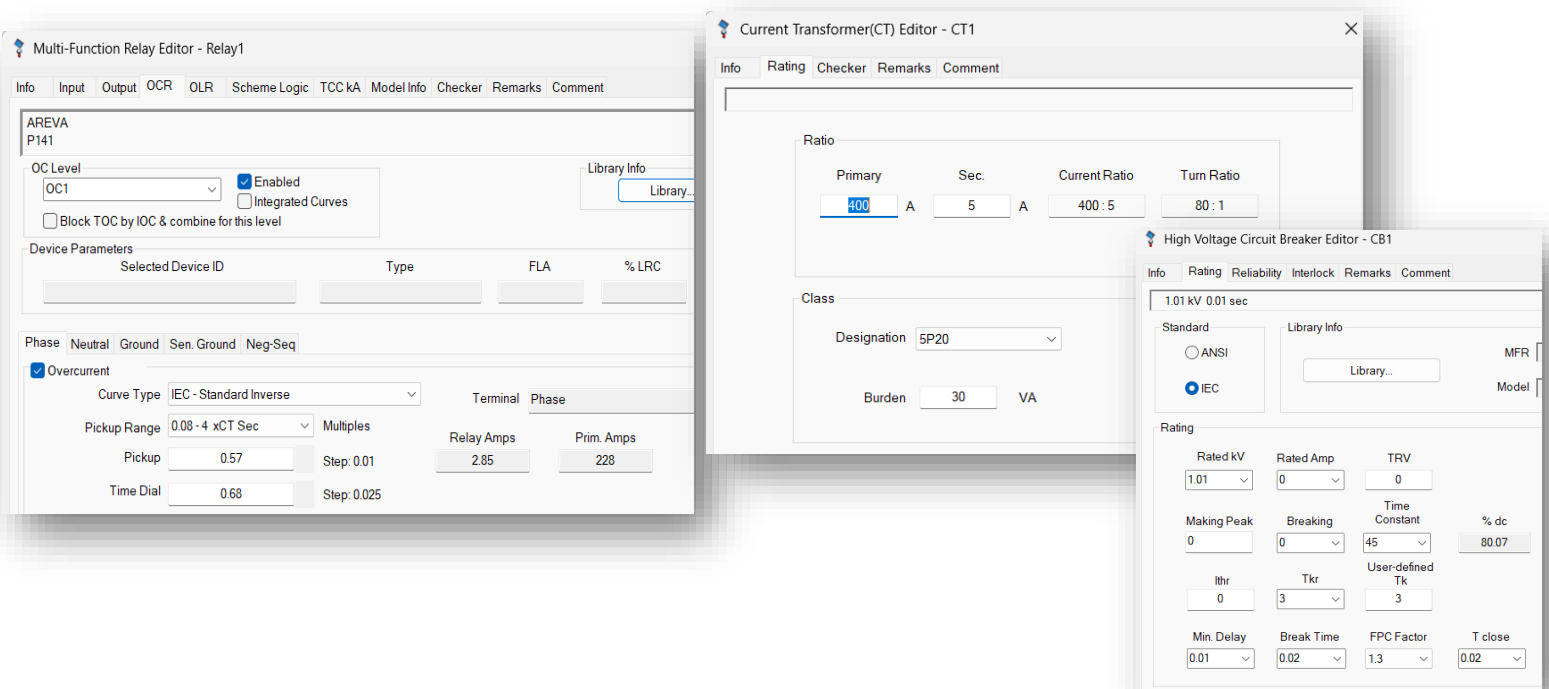
For the relay connected to the transmission line side (relay 4) calculation goes:

$t = (1.1 + 0.3) = 1.4s$ here, fault current = 14.37 KA, pickup current = 226.2 A

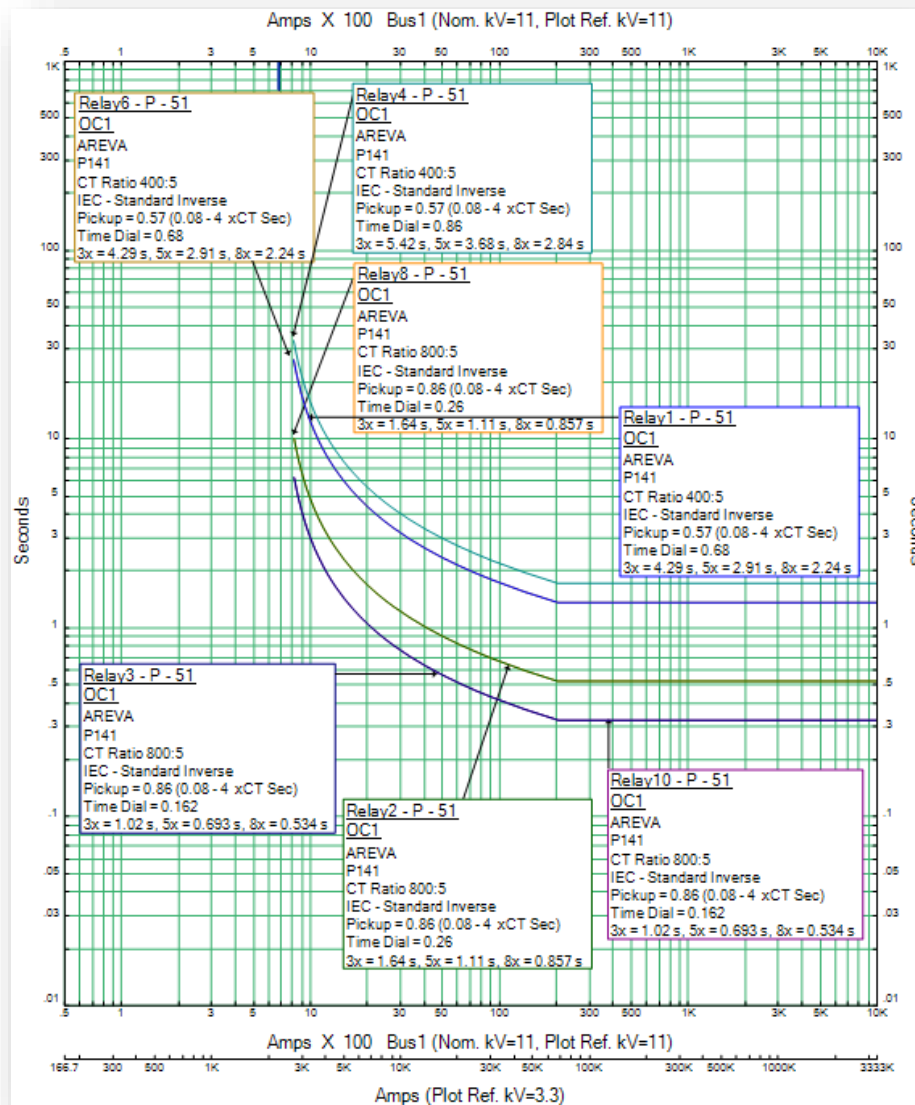
PSM = 63.52

TSM = 0.86 s

We have selected AREVA P-141 which is overload and over current relay. We have set our high voltage side CT's as 400:5 ratio and low voltage side CT's as 800:5 ratio. Our circuit breakers are 1.01 KV with 0.01 delay time and all are high voltage circuit breakers.

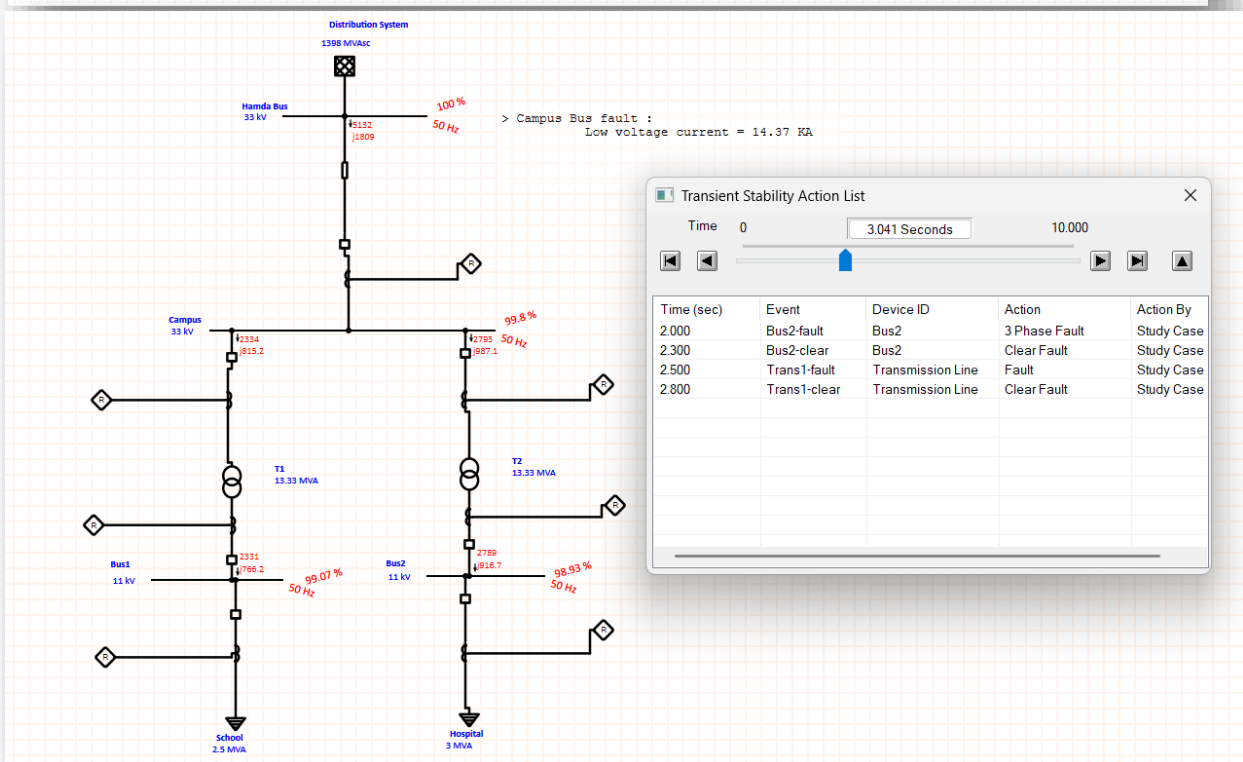
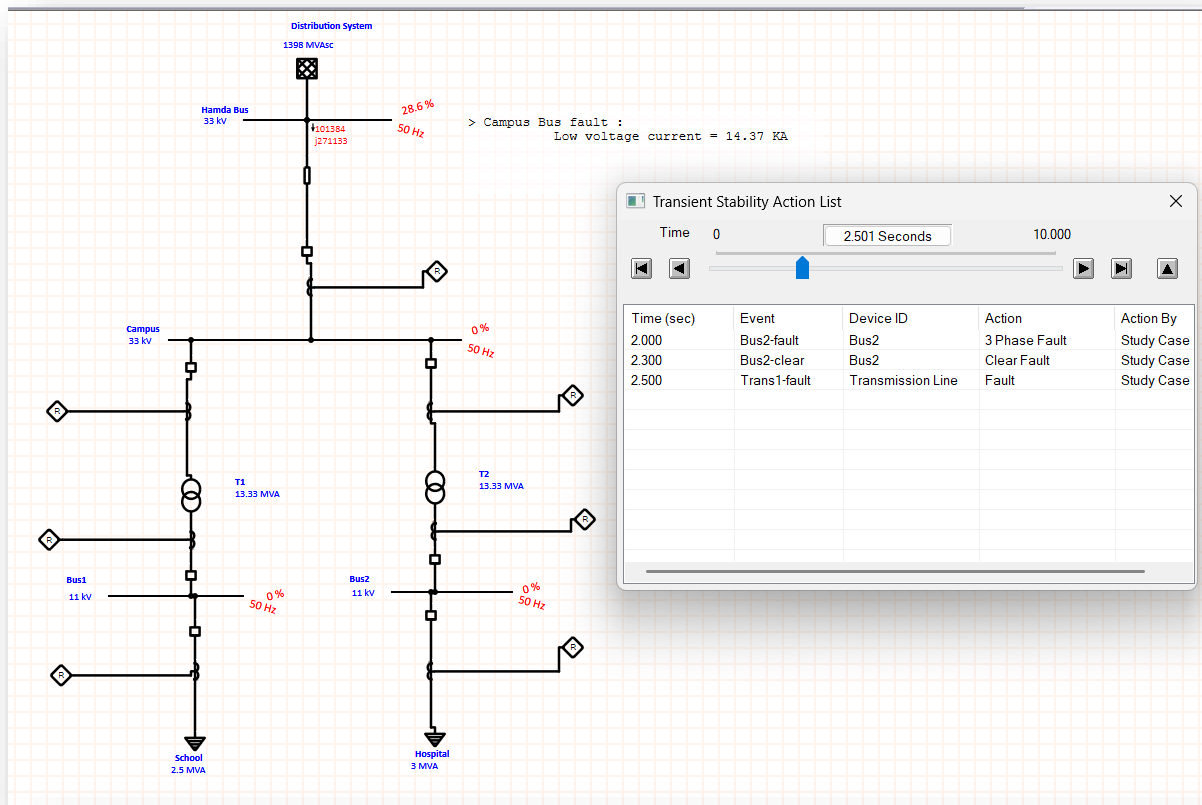


TCC Curve:

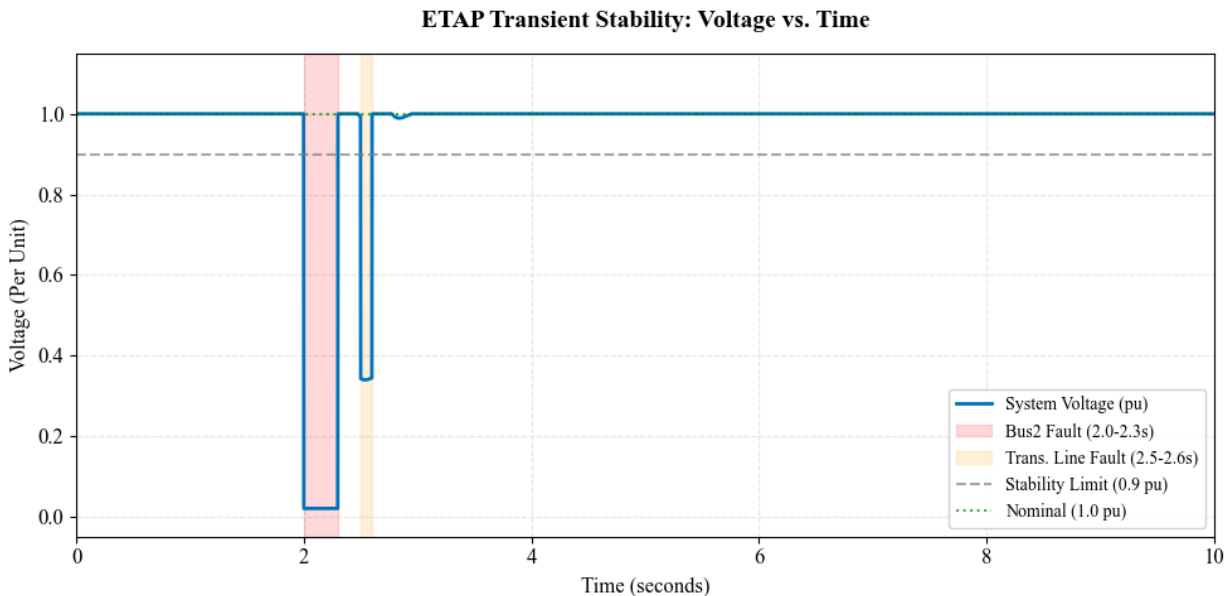


The TCC data defines a coordinated overcurrent protection scheme where six relays are set to isolate faults selectively. Each relay uses an IEC Standard Inverse curve and has a CT ratio (400:5 or 800:5) to scale current, a pickup setting (0.57 or 0.86) to start timing, and a Time Dial (0.16 to 0.88) to set its base speed. The core principle is time-graded coordination: a downstream relay (e.g., Relay3 with Time Dial 0.16) is set to operate faster than the upstream relay (e.g., Relay2 with Time Dial 0.26) protecting it. The listed operating times (like 3x=1.02s for Relay3 vs. 3x=1.64s for Relay2) create a deliberate time gap. This ensures that for a fault in Relay3's zone, it will trip in about 1 second, clearing the fault before Relay2's timer finishes at 1.64 seconds, thus preventing an unnecessary upstream outage and maintaining system reliability.

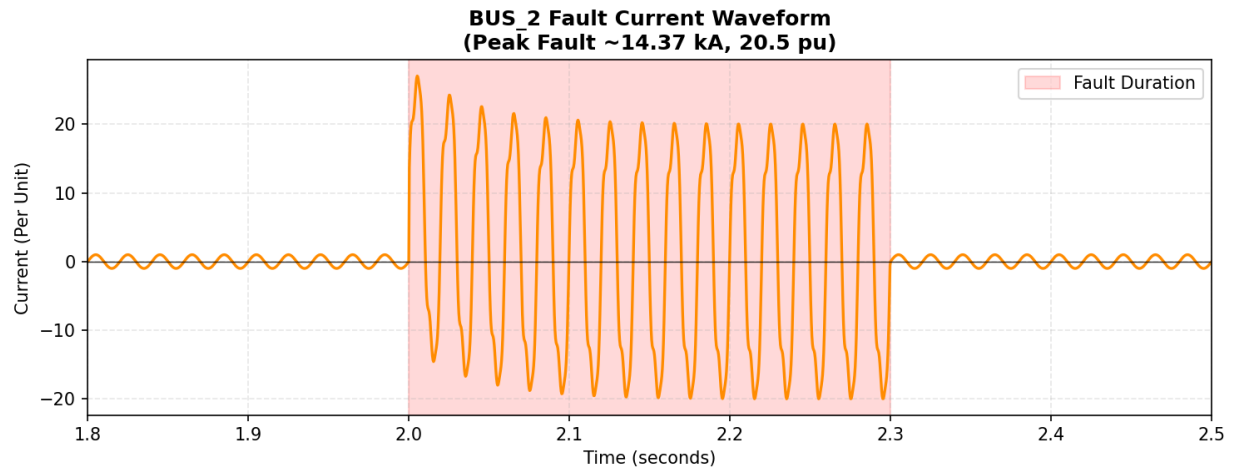
When any fault occurs on the system the percentage of the voltage drops to 0 and when the fault is cleared it again gains the ability to transfer voltage throughout the system. We can see that easily from the diagram with the study case of transmission line fault. In the case study of 4 we are showing the stability action by creating a fault in transmission line and removing fault from it also.



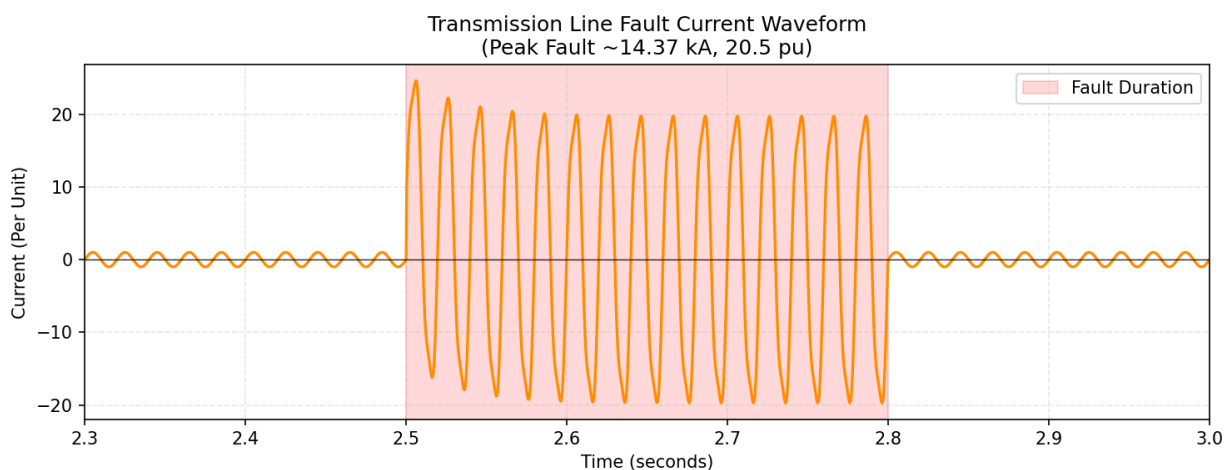
Result and Plot:



The Voltage vs. Time plot illustrates your power system's transient stability by simulating its response to staged faults from your ETAP study. During normal operation, the voltage remains steady at 1.0 per unit (pu), represented by the green dotted baseline. The first critical event is a three-phase fault at Bus2, occurring from 2.0 to 2.3 seconds, which causes a severe voltage collapse to nearly zero, as shown by the red shaded region and the blue line plummeting. Subsequently, a transmission line fault from 2.5 to 2.6 seconds creates a partial voltage dip, indicated by the orange shaded zone. Following each fault clearance, the voltage exhibits damped oscillations before recovering to its nominal value, demonstrating successful system stabilization. The plot includes a stability threshold at 0.9 pu (gray dashed line); the recovery above this line confirms that your protection scheme maintains the system within safe operational limits post-fault. This visual sequence validates that your coordinated relays and circuit breakers operated correctly to isolate the faults, thereby preserving overall grid stability. The simulation accurately reflects the event timings and actions logged in your ETAP Transient Stability Action List.



The Fault Current Waveform plot provides a detailed view of the current signal during the Bus2 fault event, which is the primary input for your overcurrent protection relays. Under normal conditions, the current is a clean, low-amplitude sinusoidal waveform at the system frequency. The onset of the three-phase short-circuit fault at 2.0 seconds triggers an instantaneous and massive surge in current magnitude, by a factor of approximately twenty. This surge corresponds to the high fault current values calculated in your ETAP study. The waveform during the fault period becomes highly distorted, featuring a pronounced DC offset a decaying exponential component caused by the asymmetrical nature of the fault inception. Superimposed on this are higher-frequency harmonics, giving the waveform a non-sinusoidal, "noisy" appearance that characterizes real-world electromagnetic transients. This distorted waveform persists for exactly 0.3 seconds, the precise duration set in your ETAP case before the protection system clears the fault. At 2.3 seconds, the current instantly returns to its normal sinusoidal state, visually confirming the circuit breaker's operation. This plot is crucial because it represents the exact electrical condition that your ANSI 51 time-overcurrent relays are designed to detect, proving that the fault current levels were sufficient to trigger the coordinated trip sequence defined in your Time-Current Characteristic (TCC) curves.



This fault current waveform plot illustrates the simulated electrical transient during a transmission line fault in your power system. The graph shows a clean, low-amplitude sinusoidal current representing normal load conditions outside the fault period. A dramatic transformation occurs within the fault duration window from 2.5 to 2.8 seconds, marked by the red shaded zone, where the current surges to a magnitude of approximately 20.5 per unit, corresponding to the 14.37 kA peak fault current from your ETAP study. The waveform during this period is heavily distorted, losing its pure sinusoidal shape due to the superimposition of a strong, exponentially decaying DC offset—a hallmark of asymmetrical fault inception. Additional harmonic distortions are visible, giving the signal a non-smooth appearance, which accurately models the electromagnetic transients in a real short-circuit event. The instantaneous return to a normal, low-magnitude sine wave at precisely 2.8 seconds confirms the successful intervention of the protection system, which has cleared the fault. This current signature is the direct input signal for your overcurrent protection relays; its high magnitude validates that the fault current exceeds the relays' pickup settings, triggering their timing logic. The defined 0.3-second fault window visually represents the precise coordination between the relay's operating time and the circuit breaker's clearing time.

Discussion:

This project successfully demonstrates the design, simulation, and validation of a time-graded, relay-based protection scheme for a sample power system using ETAP software. The core achievement is the establishment of a coordinated overcurrent protection strategy employing IEC Standard Inverse characteristic curves. By calculating appropriate plug settings (PS) and time multiplier settings (TMS) for each relay based on fault current levels and a predetermined time grading interval of 0.3 seconds, the scheme ensures selective fault isolation. The Time-Current Characteristic (TCC) curve visually confirms this coordination, where downstream relays operate faster than their upstream counterparts, thereby minimizing outage scope and maintaining system reliability. The workflow was rigorously validated through ETAP simulations, including load flow analysis to confirm normal operating conditions, short-circuit analysis to determine fault currents, and transient stability studies. The stability check and resultant plots (Voltage vs. Time and Fault Current Waveforms) provide compelling evidence that the designed scheme effectively clears staged three-phase and transmission line faults within specified times, allows for voltage recovery, and maintains system stability above the critical threshold, thereby protecting equipment and ensuring continuity of service.

Limitations and Future Improvements

The presented design, while effective for the defined scenario, incorporates several simplifications that point to avenues for future enhancement. Key limitations include the use of fixed, pre-calculated settings based on a limited set of fault studies, which may not adapt optimally to varying system configurations, loading conditions, or unconventional fault types (e.g., high-impedance faults). The protection scheme relies solely on independent time-

overcurrent principles without communication between relays, which can limit tripping speed for certain faults and complicate coordination in more complex, multi-source networks. Furthermore, the simulation uses standard relay models and assumes ideal circuit breaker operation.

Future improvements would involve transitioning from a static to an adaptive protection system. This could be achieved by integrating digital relays capable of receiving real-time measurements via communication protocols (e.g., IEC 61850 GOOSE), enabling faster and more selective protection schemes like differential protection or distance protection with communication-aided tripping. Incorporating advanced relay models with more nuanced characteristics and modeling the dynamic behavior of sources like inverter-based resources (e.g., solar PV, wind farms) would increase simulation fidelity. Finally, expanding the study to include a comprehensive analysis of system resilience under N-1 contingency conditions and the impact of distributed generation would make the protection scheme more robust and applicable to modern power grids.